

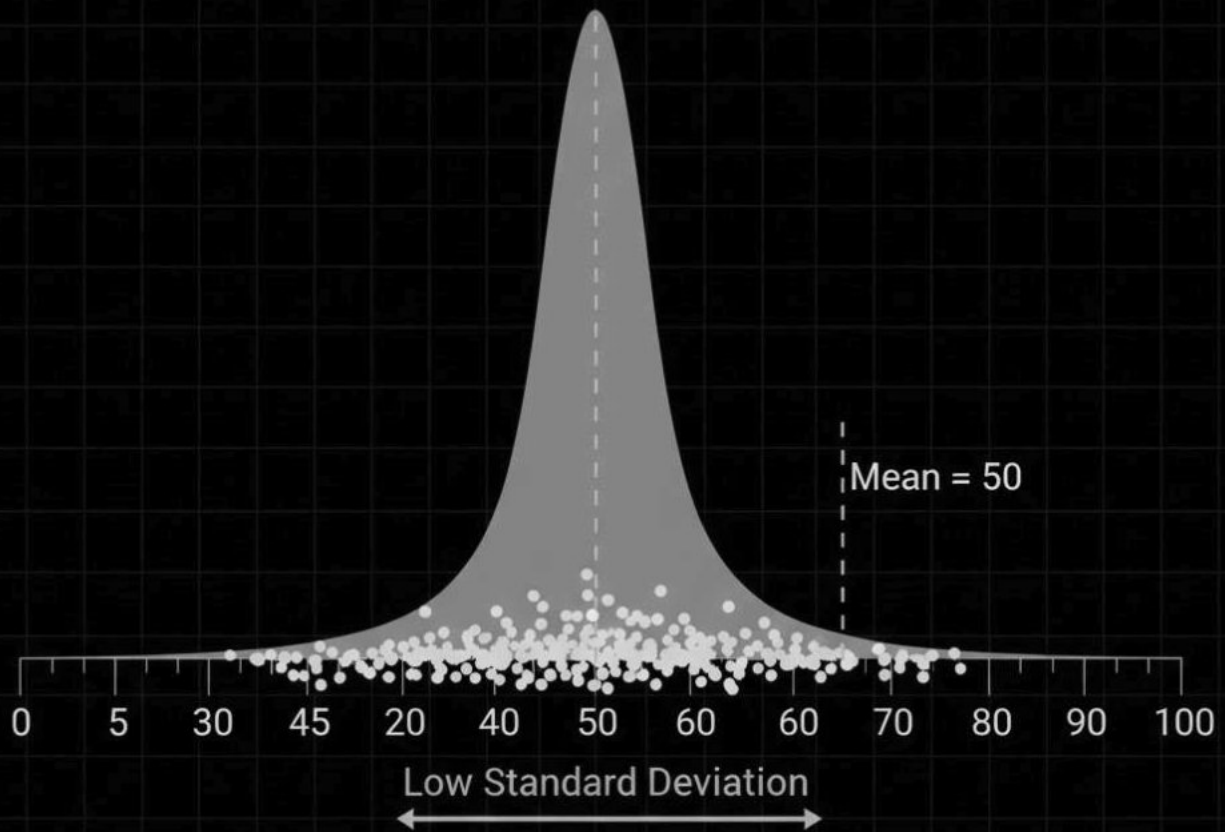
Measures Of Dispersion

Measures of **Dispersion** (or variability, scatter, or spread) are statistical values that describe **how spread out** or scattered the data points are in a dataset. They tell you the degree to which individual items in a distribution vary from each other and from the central value (like the mean or median).

- **Low Dispersion:** Data points are closely clustered around the mean, indicating high consistency.
- **High Dispersion:** Data points are widely scattered, indicating low consistency.

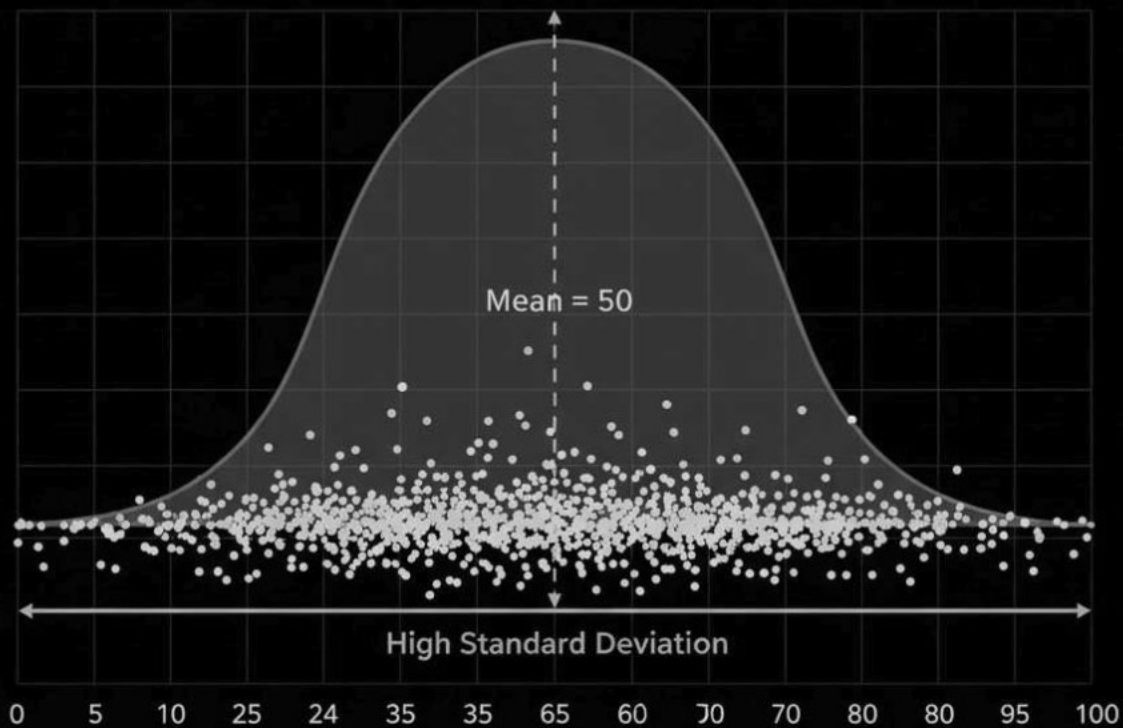
Low Dispersion

Data points are clustered tightly around the mean



High Dispersion

Data points are widely scattered from the mean



While measures of central tendency (mean, median, mode) tell you where the **center** of the data is, measures of dispersion tell you about the **reliability** of that center.

σ

Interquartile Range (IQR)

The Interquartile Range is a measure of dispersion that focuses only on the **middle 50%** of the data. It is highly valued because it is **not affected by extreme values** (outliers) in

the data set.

Definition

The Interquartile Range (**IQR**) is the difference between the **Third Quartile** (Q_3) and the **First Quartile** (Q_1).

- **Quartiles** are the values that divide an ordered dataset into **four equal parts** (each containing 25% of the data).
 - Q_1 (**First Quartile / Lower Quartile**): The value below which 25% of the data lies. It is essentially the median of the lower half of the data.
 - Q_2 (**Second Quartile / Median**): The value below which 50% of the data lies.
 - Q_3 (**Third Quartile / Upper Quartile**): The value below which 75% of the data lies. It is essentially the median of the upper half of the data.

Formula

$$\text{IQR} = Q_3 - Q_1$$

Example: Ungrouped Data

- **Data:** 5, 7, 10, 12, 15, 18, 20, 22, 25 ($N = 9$)
 1. **Order Data:** 5, 7, 10, 12, 15, 18, 20, 22, 25
 2. **Find Q_2 (Median):** The middle value is the $(9 + 1)/2 = 5$ th term, which is **15**.
 3. **Find Q_1 :** This is the median of the lower half (5, 7, 10, 12). The average of the two middle terms is $(7 + 10)/2 = \mathbf{8.5}$.
 4. **Find Q_3 :** This is the median of the upper half (18, 20, 22, 25). The average of the two middle terms is $(20 + 22)/2 = \mathbf{21}$.
 5. **Calculate IQR:** $Q = Q_3 - Q_1 = 21 - 8.5 = \mathbf{12.5}$.
- **Interpretation:** The middle 50% of the data values are spread over a range of 12.5 units.



Variance (² or ²)

Variance is the **most common** and mathematically robust measure of dispersion. It measures the **average of the squared differences** from the mean.

Definition

Variance is the average of the squared deviations of each observation from the **Arithmetic Mean** ($\bar{x}$ or μ) of the dataset.

- **Why Square the Differences?**

1. To ensure that the deviations are **positive**. If we just sum the raw deviations from the mean, the total sum will always be zero.
2. To give **greater weight** to larger deviations, meaning observations that are far away from the mean have a much greater impact on the spread.

- **Population vs. Sample:**

- **Population Variance** (σ^2 - sigma squared): Calculated when the data represents the *entire* group of interest. The denominator is N (total observations).
- **Sample Variance** (s^2): Calculated when the data is a *subset* used to estimate the characteristics of a larger population. The denominator is $n - 1$ (degrees of freedom) to provide a better, **unbiased estimate** of the population variance.

For An Individual Series (Ungrouped Data)

An individual series is a raw list of observations (x_i).

Formula (Population Variance σ^2):

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

Where:

- x_i = Each individual observation
- $\bar{x}$ = Population Mean ($\frac{\sum x_i}{N}$)
- N = Total number of observations

Formula (Sample Variance s^2):

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Where:

- $\bar{x}$ = Sample Mean ($\frac{\sum x_i}{N}$)
- N = Sample size

Example (Using Population Variance):

- **Data (x_i):** 2, 4, 6, 8, 10 ($N = 5$)

1. **Calculate Mean ($\bar{x}$):** $= \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$

2. **Calculate Squared Deviations ($(x_i - \bar{x})^2$):**

- $(2 - 6)^2 = (-4)^2 = 16$
- $(4 - 6)^2 = (-2)^2 = 4$
- $(6 - 6)^2 = (0)^2 = 0$
- $(8 - 6)^2 = (2)^2 = 4$
- $(10 - 6)^2 = (4)^2 = 16$

3. **Sum of Squared Deviations ($\sum (x_i - \bar{x})^2$):** $16 + 4 + 0 + 4 + 16 = 40$

4. **Calculate Variance (σ^2):** $\sigma^2 = \frac{40}{5} = 8$

For A Frequency Distribution (Discrete or Continuous)

When data is presented with frequencies (f), the deviation of each value is weighted by how often it occurs.

Formula (Population Variance σ^2):

$$\sigma^2 = \frac{\sum f_i(x_i - \bar{x})^2}{N}$$

Where:

- f_i = Frequency of the observation
- x_i = Observation value (or **Midpoint** for continuous data)
- $\bar{x}$ = Population Mean
- N = Total Frequency ($\sum f_i$)

Formula (Sample Variance s^2):

$$s^2 = \frac{\sum f_i(x_i - \bar{x})^2}{n - 1}$$

Where:

- $\bar{x}$ = Sample Mean
- N = Total Sample Frequency ($\sum f_i$)

Example (Using Population Variance for Discrete Data):

x_i	f_i	$f_i x_i$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$f_i (x_i - \bar{x})^2$
10	2	20	-20	400	800
20	3	60	-10	100	300
30	5	150	0	0	0
40	4	160	10	100	400
50	1	50	20	400	400
Total	$N = 15$	$\sum f_i x_i = 440$	-	-	$\sum f_i (x_i - \bar{x})^2 = 1900$

1. **Calculate Mean ($\bar{x}$):** $\bar{x} = \frac{\sum f_i x_i}{N} = \frac{440}{15} \approx 29.33$.
2. **Calculate Variance (σ^2):** $\sigma^2 = \frac{1900}{15} \approx 126.67$.

Standard Deviation

- **Advanced Note (Standard Deviation):** The **Standard Deviation** (or σ) is simply the **square root of the variance**. It is often preferred because it is expressed in the same units as the original data, making it easier to interpret the spread (e.g., if the mean score is 30, a standard deviation of 5 means a typical score is 5 points away from the mean).